

7 Animals and plants can be genetically modified.

- (a) Recombinant human erythropoietin (rHE) is a protein drug that can be given to patients with low haemoglobin concentrations.
- (i) Sheep have been genetically modified to produce rHE in their milk.

Explain how a sheep could be genetically modified to produce rHE in its milk.

(4)

This image shows a full page of white paper with horizontal dashed lines, typical of primary-ruled notebook paper. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings present.

- (ii) In a clinical study, 4382 patients were treated with rHE. They showed a mean increase in blood haemoglobin from 80 g dm^{-3} to 142 g dm^{-3} .

Calculate the percentage increase in the mean blood haemoglobin concentration as a result of rHE treatment.

(1)

Answer %

- (iii) Some patients receiving rHE develop anti-rHE antibodies.

Describe how these antibodies may be produced.

(4)

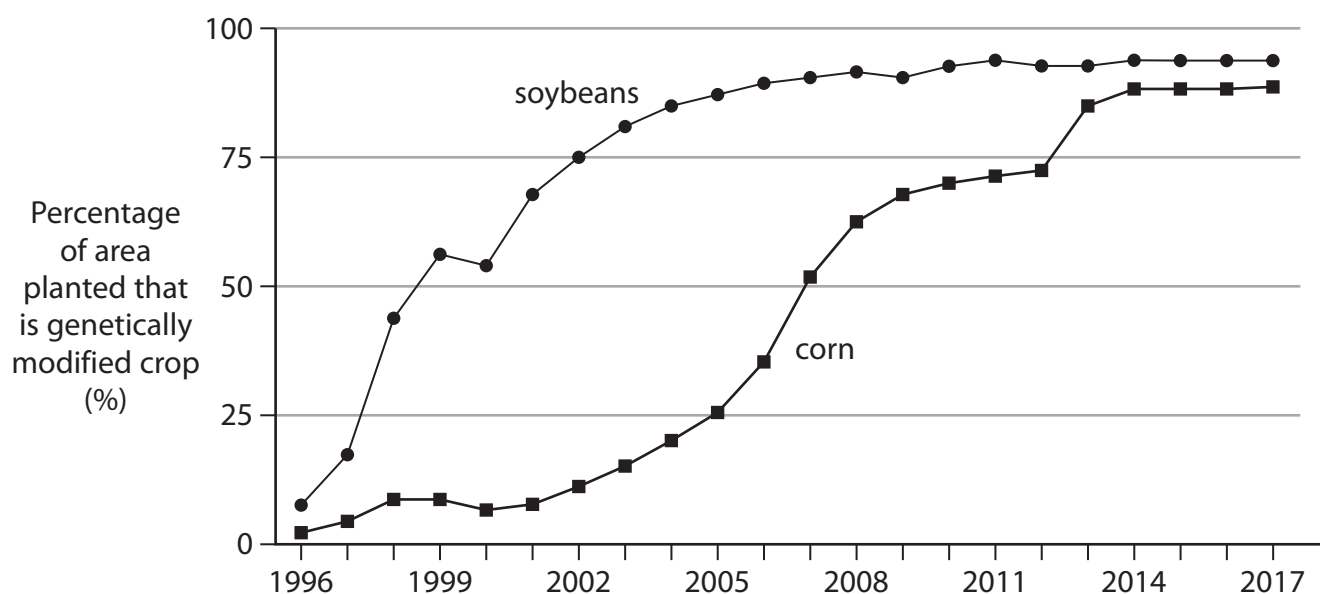
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- (b) Advances in genetic engineering have allowed for precise control of the genetic changes introduced into plants.

The table describes some examples of the use of genetic modification in plants.

Trait	Plant	Details
herbicide tolerance	soybean	glyphosphate herbicide tolerance using a gene from the bacterium <i>A. tumefaciens</i>
insect resistance	corn	resistance to European corn borer using a gene from <i>B. thuringiensis</i>

The graph shows the increase in the use of genetically modified crops in the United States from 1996 to 2017.



Discuss the risks and benefits of genetic modification in valuable agricultural plants.

Use all the data and your own knowledge to support your answer.

(6)

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(Total for Question 7 = 15 marks)